Experiment 1

Implementation of Data Encryption Standard (DES)

Aim

To implement the **Data Encryption Standard (DES)** algorithm in Python for encrypting and decrypting text using a secret key, ensuring confidentiality of data.

Description

The **Data Encryption Standard (DES)** is a symmetric-key block cipher that encrypts data using the same secret key for both encryption and decryption. It operates on **64-bit blocks** of plaintext and uses a **56-bit effective key** (stored as 64 bits with 8 parity bits). DES follows the **Feistel network** structure and performs **16 rounds** of substitution and permutation to produce the ciphertext. During decryption, the same process is followed, but the round keys are applied in reverse order.

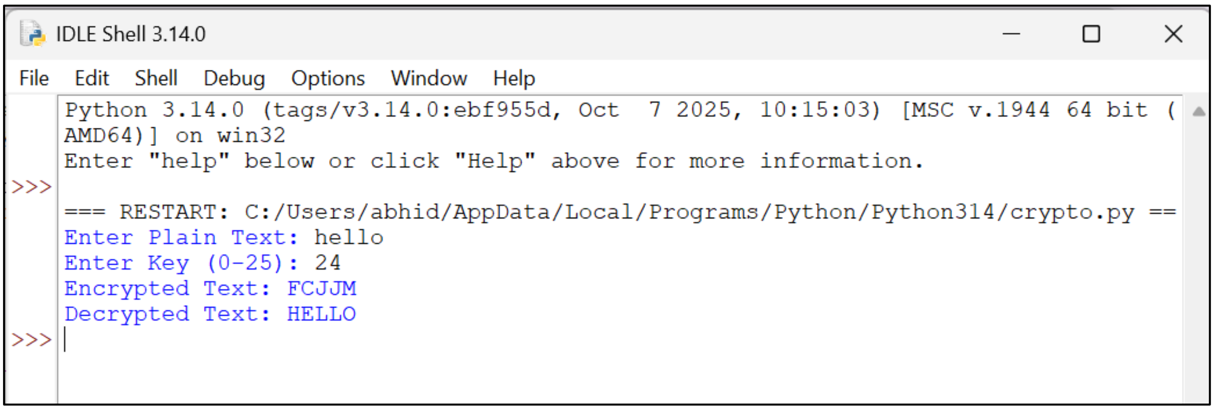
Algorithm

1. Start the program.
2. Import the required DES library.
3. Read the plaintext from the user.
4. Read the secret key (8 characters).
5. Convert the key into bytes.
6. Create a DES cipher object.
7. Pad the plaintext to make its length a multiple of 8 bytes.
8. Encrypt the plaintext using the DES cipher.
9. Display the encrypted text.
10. Decrypt the ciphertext using the same key.
11. Remove the padding from the decrypted text.
12. Display the decrypted text.
13. Stop the program.

Program

```
from Crypto.Cipher import DES
from Crypto.Util.Padding import pad, unpadimport base64
key = input("Enter 8-character Key: ").encode()
plaintext = input("Enter Plain Text: ").encode()
cipher = DES.new(key, DES.MODE_ECB)
encrypted = cipher.encrypt(pad(plaintext, DES.block_size))
print("Encrypted Text:", base64.b64encode(encrypted).decode())
decrypted = unpad(cipher.decrypt(encrypted), DES.block_size)
print("Decrypted Text:", decrypted.decode())
```

Output:



```
IDLE Shell 3.14.0
File  Edit  Shell  Debug  Options  Window  Help
Python 3.14.0 (tags/v3.14.0:ebf955d, Oct  7 2025, 10:15:03) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
=== RESTART: C:/Users/abhid/AppData/Local/Programs/Python/Python314/crypto.py ==
Enter Plain Text: hello
Enter Key (0-25): 24
Encrypted Text: FCJJM
Decrypted Text: HELLO
>>>
```